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ORIGINAL RESEARCH

Left Atrial Mechanical Dysfunction and the Risk for Ischemic Stroke in People Without Prevalent Atrial Fibrillation or Stroke

A Prospective Cohort Study

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Background: Atrial myopathy—characterized by changes in left atrial function and size—may precede and promote atrial fibrillation (AF) and cardiac thromboembolism. In people without prior AF or stroke, whether analysis of left atrial function and size can improve ischemic stroke prediction is unknown.

Objective: To evaluate the association of echocardiographic left atrial function (reservoir, conduit, and contractile strain) and left atrial size (left atrial volume index) with ischemic stroke and determine whether these measures can improve the stroke prediction achieved by CHA_2DS_2-VASc score variables.

Design: Prospective cohort study.

Setting: ARIC (Atherosclerosis Risk in Communities) study.

Participants: 4917 ARIC participants without prevalent stroke or AF.

Measurements: Ischemic stroke events (2011 to 2019) were adjudicated by physicians. Left atrial strain was measured using speckle-tracking echocardiography.

Results: Over 5 years, the cumulative incidences of ischemic stroke in the lowest quintiles of left atrial reservoir, conduit, and contractile strain were 2.99% (95% CI, 1.89% to 4.09%), 3.18% (CI, 2.14% to 4.22%), and 2.15% (CI, 1.09% to 3.21%), respectively, and that of severe left atrial enlargement was 1.99% (CI, 0.23% to 3.75%). On the basis of the Akaike information criterion, left atrial reservoir strain plus CHA_2DS_2-VASc variables was the best predictive model. With the addition of left atrial reservoir strain to CHA_2DS_2-VASc variables, 11.6% of the 112 participants with stroke after 5 years were reclassified to higher risk categories and 1.8% to lower risk categories. Among the 4805 participants who did not develop stroke, 12.2% were reclassified to lower and 12.7% to higher risk categories. Decision curve analysis showed a predicted net benefit of 1.34 per 1000 people at a 5-year risk threshold of 5%.

Limitation: Underascertainment of subclinical AF.

Conclusion: In people without prior AF or stroke, when added to CHA_2DS_2-VASc variables, left atrial reservoir strain improves stroke prediction and yields a predicted net benefit, as shown by decision curve analysis.

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Atrial fibrillation (AF) is a serious public health problem because of its increasing prevalence in the aging population and its association with elevated risks for cardiovascular outcomes, such as cardiac thromboembolism and stroke (1). Mounting evidence suggests that an intrinsically prothrombotic atrial myopathy (2–4)—characterized by changes in atrial mechanical function and size—precedes and promotes development of AF (3, 5) (**Supplement Figure 1**, available at Annals.org). Evaluating measures of left atrial function and size may have utility in enhancing prediction of cardiac embolism and stroke earlier in a patient's disease course before development of AF.

Greater left atrial size has traditionally been considered to be a marker of atrial myopathy and stroke risk (6). Of note, left atrial mechanical dysfunction can be detected before left atrial enlargement through analysis of left atrial strain (7–11). Lower left atrial strain, which reflects impaired tissue compliance or contractility, has recently been associated with ischemic stroke in small cohorts of patients with prevalent AF or stroke (12–15). However, data from prospective, community-based cohorts in people without AF or prior ischemic stroke are limited. Furthermore, whether considering measures of left atrial function and size would improve ischemic stroke prediction in such people is unknown.

Hence, the present study had 2 primary aims. The first was to evaluate the association of measures of left atrial mechanical function (left atrial reservoir, conduit, and contractile strain) and left atrial size (left atrial volume index [LAVI]) with ischemic stroke in people without prior AF or stroke. The second was to quantify the extent to which these left atrial measures improved ischemic stroke prediction above that achieved by variables from the CHA_2DS_2-VASc score, a validated prediction tool for cardioembolic stroke (16). Data were obtained from the ARIC (Atherosclerosis Risk in Communities) study, a prospective, community-based cohort study of Black and White adults in the United States.

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METHODS

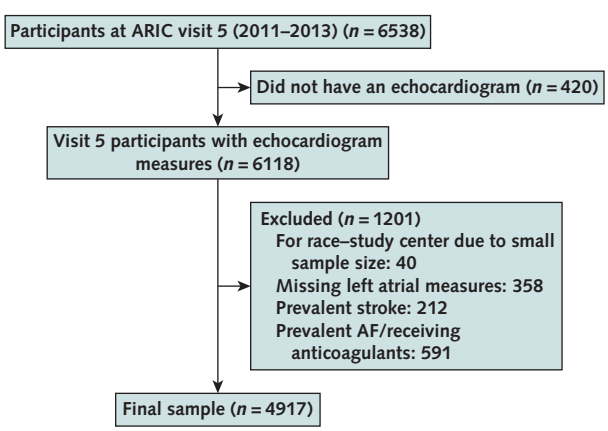
Study Population

The ARIC study recruited 15 792 people from 4 U.S. communities (Washington County, Maryland; Forsyth County, North Carolina; Jackson, Mississippi; and suburban Minneapolis, Minnesota). Since the baseline examination (1987 to 1989), 7 study visits have been completed. Between study visits, annual (semiannual since 2012) telephone interviews with participants or proxies have been done to assess health status and ascertain cardiovascular events and deaths. In addition, regional hospitalization records in each of the ARIC communities are systematically reviewed to ascertain cardiovascular events and deaths. Finally, there is linkage to the National Death Index.

For the present analysis, participants from visit 5 (2011 to 2013) were followed for events through the end of 2019 (2017 for Jackson). As of visit 5, a total of 1917 participants had been lost to follow-up or withdrawn from the study and 4587 were deceased. Of the remaining 9288 participants still active in the study, 6538 attended visit 5. In 2012, 91% of participants or their proxies participated in annual or semiannual telephone calls to collect follow-up data. In 2019, the last year of our analysis, the contact rate was 81%.

Two-dimensional echocardiograms were obtained for 6118 of visit 5 attendees. Participants were excluded from the present analysis for missing measures of left atrial strain (n = 285) or missing LAVI (n = 73). Because of small sample size (n = 40), we excluded participants who were not White or Black from all study sites and those who were not White from Minneapolis and Washington County. Finally, we excluded participants with prevalent stroke (n = 212) and those with prevalent AF or anticoagulant use (n = 591), resulting in a final analytic cohort of 4917 participants (**Figure 1**). Additional details on ARIC study design, outcome ascertainment, and population statistics have been previously described (4, 17–20) and are provided in the **Supplement Methods** (available at Annals.org).

Figure 1. Study flow diagram.



ARIC study, 2011–2019. Missing echocardiograms were due to participants opting out of the examination. Missing left atrial strain measures were due to suboptimal image quality. AF = atrial fibrillation; ARIC = Atherosclerosis Risk in Communities.

Echocardiography

Echocardiography protocols in ARIC have been previously described (21, 22). Left atrial function has 3 phases in the cardiac cycle: reservoir, conduit, and contractile. In the reservoir phase, atrial filling occurs during ventricular systole as the mitral valve closes and the atrium expands with inflow of blood from the pulmonary veins. The conduit phase occurs during early ventricular diastole when the mitral valve opens and passive ventricular filling begins. The contractile phase occurs in late ventricular diastole when the atrium contracts, augmenting ventricular filling.

Left atrial maximal volume was calculated just before mitral valve opening from the apical 2- and 4-chamber views using the biplane method of disks. We calculated LAVI by indexing to body surface area.

Left atrial strain was calculated using a 2-dimensional speckle-tracking software with an autostrain algorithm (QLAB Advanced Quantification Software 123.0; Philips Ultrasound). The atrial cycle was measured between R waves (R-R gating) from successive QRS complexes (**Supplement Figure 2**, available at Annals.org). The left atrial wall was traced, and the percentage change in length (strain) over the course of each phase of the atrial cycle was calculated by tracking acoustic artifacts or speckles. The absolute value of the peak strain was then reported for each phase of the atrial cycle. Of note, left atrial conduit and contractile strain constitute left atrial reservoir strain (**Supplement Figure 2**).

Covariates

The covariates included in this study—age, race–study center, sex, heart failure (23), hypertension (24, 25), diabetes (25, 26), myocardial infarction (27, 28), peripheral artery disease (28), and incident AF (29, 30)—were derived from data obtained during the ARIC study as previously described. With the exception of race–study center and incident AF, the covariates constitute the CHA_2DS_2-VASc score. Incident AF was updated to the end of follow-up, and the remaining covariates were obtained at visit 5. Additional details are provided in the **Supplement Methods**.

Stroke Adjudication and Classification

Strokes were adjudicated using criteria from the National Survey of Stroke by the National Institute of Neurological Disorders and Stroke (31). In the ARIC study, strokes were classified as definite or probable thrombotic stroke, definite or probable noncarotid embolic stroke, definite or probable subarachnoid hemorrhage, definite or probable brain hemorrhage, possible stroke of undetermined type, or out-of-hospital fatal stroke. Definite thrombotic strokes were subclassified as definite thrombotic lacunar or definite thrombotic nonlacunar stroke. Our primary outcome was ischemic stroke, which included all definite or probable noncarotid embolic and thrombotic strokes.

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Details on stroke classification criteria in the ARIC study can be found in the **Supplement Methods** and the ARIC study protocol manual 3 (www2.cscc.unc.edu/aric) (18).

Statistical Analysis

Baseline characteristics are presented as numbers (percentages) for categorical variables and means (SDs) for continuous variables.

The association of each left atrial measure with ischemic stroke was visualized using restricted cubic splines (graphed as adjusted incidence rates) and cumulative incidence curves, each adjusted for age, sex, race–study center, heart failure, hypertension, myocardial infarction, peripheral artery disease, diabetes, and LAVI (for left atrial strain parameters) or left atrial reservoir strain (for LAVI). Cumulative incidence curves additionally accounted for the competing risk for death. For restricted cubic splines, the number of knots (3) was selected on the basis of the lowest Akaike information criterion (AIC), compared with 5 and 7 knots. The knots were placed at the following percentiles of each left atrial parameter: 10%, 50%, and 90%. Poisson models and likelihood ratio tests evaluated the overall association and linearity of cubic splines.

For cumulative incidence curves, left atrial strain measures were categorized by quintile. We categorized LAVI on the basis of clinical cutoffs as recommended by the American Society of Echocardiography (32).

We used Cox proportional hazards models to estimate hazard ratios and 95% CIs of each left atrial measure for ischemic stroke. Model prediction error was estimated by AIC: Models with lower AIC were better at predicting the outcome than models with higher AIC. The benchmark model (model 0) included CHA_2DS_2-VASc variables (age, sex, heart failure, diabetes, peripheral artery disease, and myocardial infarction). There were 4 versions of model 1 (A to D), each adding a left atrial measure to model 0. Models 2, 3, and 4 each added to model 0 the left atrial measure associated with the best model 1 AIC and another left atrial measure in ascending order of its respective model 1 AIC. Finally, to further evaluate which left atrial parameter was the strongest independent predictor of ischemic stroke, we generated a parsimonious model with stepwise backward selection (33). Additional details are provided in the **Supplement Methods**. Left atrial measures were modeled as continuous variables based on clinically relevant intervals (per 5–percentage point decrement for left atrial strain [34] and per 5-mL/m^2 increment for LAVI).

We did several secondary analyses. First, we evaluated the association of left atrial measures with ischemic stroke subtypes—thrombotic stroke (lacunar and nonlacunar stroke) and noncarotid cardioembolic stroke. Second, because the range of values and SDs was different for the 4 left atrial measures (SD largest for left atrial reservoir strain), modeling per 1-SD decrement (left atrial strain) and 1-SD increment (LAVI) was done to facilitate comparisons across left atrial measures.

We did several sensitivity analyses. First, each version of model 1 was additionally adjusted for covariates not included in the benchmark model, such as incident AF, race–study center, and medication use associated with reversal of atrial myopathy (35–38). Second, to evaluate the association of left atrial strain measures with ischemic stroke in participants with normal left atrial size as defined by the American Society of Echocardiography (32), we excluded participants with a LAVI of 34 mL/m^2 or higher. Third, to further ensure that the study findings were independent of AF or anticoagulation, we excluded participants with incident AF or subsequent anticoagulant use after visit 5. Fourth, to account for the competing risk for death during follow-up, we calculated subdistribution hazard ratios (95% CIs) using Fine–Gray models for the analysis of time-to-event outcomes. Fifth, to account for bias that might have been introduced by nonattendance at visit 5, we used inverse probability weighting. Additional details are in the **Supplement Methods**.

Improvement in 5-year reclassification of ischemic stroke risk by the model with the lowest AIC compared with the benchmark model (model 0) was evaluated using risk classification tables with the following risk categories: higher than 5%, 2.5% to 5%, and lower than 2.5%. Categorical net reclassification improvement and integrated discrimination improvement were also calculated.

To evaluate the potential for clinical application, we generated decision curves (39) to plot the predicted net benefit (the total true-positive classification minus the total false-positive classification weighted by the odds of the absolute risk threshold) over a range of 5-year absolute risk thresholds for model 0 and the model with the lowest AIC, as well as intervention-for-all and intervention-for-none strategies.

Statistical analysis of data was done using SAS, version 9.4 (SAS Institute) and STATA 14.0 (StataCorp).

Role of the Funding Source

The ARIC study has been funded in whole or in part with federal funds from the National Heart, Lung, and Blood Institute of the National Institutes of Health. The funding sources had no role in the design, conduct, or reporting of the present analysis.

RESULTS

Study Population

Of the 4917 participants (mean age, 75.2 years; 59.6% women; 21.0% Black), 176 (3.6%) developed ischemic stroke over a mean follow-up time of 6.6 years (SD, 1.8). In general, participants with severe left atrial enlargement and those in the lowest quintile of left atrial strain (conduit, reservoir, and contractile) had a higher prevalence of CHA_2DS_2-VASc variables than those with normal left atrial size and those in the highest quintile of left atrial strain (**Table 1**). Although the contact rate with participants in 2019 was 81%, because of ARIC surveillance only 211 of 4917 participants (4.3%) had incomplete follow-up data. In general, baseline characteristics were similar between participants with incomplete follow-up data and those who completed follow-up except that those with incomplete follow-up data were significantly more likely to be White (84.8% vs. 78.8%) (**Supplement Table 1**, available at Annals.org).

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Table 1. Baseline Participant Characteristics at Visit 5

Variable*	Left Atrial Reservoir Strain				
	Quintile 1 (≤26%)	**Quintile 2 (>26% to ≤30%)**	**Quintile 3 (>30% to ≤34%)**	**Quintile 4 (>34% to ≤39%)**	**Quintile 5 (>39%)**
Participants, *n*	875	978	1022	1006	1036
Mean age (SD), *y*	77.3 (5.3)	75.7 (5.1)	75.1 (4.8)	74.4 (4.8)	74.0 (4.6)
Female sex, %	61.3	60.1	60.6	57.1	59.3
Black race, %	24.2	25.1	21.9	19.0	15.4
Medical conditions, %					
Heart failure	13.3	4.4	3.1	3.2	2.8
Hypertension	79.5	77.2	71.9	68.9	66.3
Myocardial infarction	15.9	9.6	5.5	5.4	3.6
Peripheral artery disease	16.5	8.7	7.0	7.5	6.1
Diabetes	36.7	32.7	27.8	28.7	24.3
Medications, %					
Aspirin	73.1	71.3	68.2	67.2	64.1
ACEI/ARB	49.7	49.1	45.2	41.6	40.2
Potassium-sparing diuretics†	2.7	1.3	1.8	0.80	1.5
	Left Atrial Conduit Strain				
	Quintile 1 (<10%)	**Quintile 2 (≤10% to <13%)**	**Quintile 3 (≥13% to <16%)**	**Quintile 4 (≥16% to ≤19%)**	**Quintile 5 (>19%)**
Participants, *n*	1047	952	1016	832	1070
Mean age (SD), *y*	77.3 (5.3)	75.9 (5.0)	75.0 (4.7)	74.4 (4.8)	73.5 (4.5)
Female sex, %	58.2	59.1	58.9	60.2	61.7
Black race, %	22.8	23.0	21.9	17.3	19.3
Medical conditions, %					
Heart failure	10.7	5.0	3.8	3.3	2.4
Hypertension	80.3	77.7	73.0	69.8	61.9
Myocardial infarction	15.0	7.9	6.8	4.3	4.0
Peripheral artery disease	14.4	11.5	7.3	5.5	5.4
Diabetes	37.0	34.8	30.5	23.9	22.3
Medications, %					
Aspirin	71.0	70.5	70.4	68.0	63.5
ACEI/ARB	51.1	49.9	45.7	42.0	36.3
Potassium-sparing diuretics†	2.1	1.8	1.7	1.1	1.3
	Left Atrial Contractile Strain				
	Quintile 1 (≤13%)	**Quintile 2 (>13% to <17%)**	**Quintile 3 (≥17% to ≤19%)**	**Quintile 4 (>19% to ≤22%)**	**Quintile 5 (>22%)**
Participants, *n*	892	1250	769	929	1077
Mean age (SD), *y*	75.8 (5.2)	75.2 (5.1)	74.9 (4.9)	75.1 (5.1)	75.2 (4.8)
Female sex, %	61.9	62.3	57.9	59.0	56.4
Black race, %	24.1	24.0	20.9	18.9	16.7
Medical conditions, %					
Heart failure	8.7	5.4	3.6	3.4	4.4
Hypertension	73.8	71.2	74.4	70.5	73.9
Myocardial infarction	11.4	8.2	6.5	6.7	5.9
Peripheral artery disease	11.4	9.4	7.7	8.1	7.9
Diabetes	30.2	29.5	31.1	27.7	30.8
Medications, %					
Aspirin	71.5	69.6	66.2	66.5	68.6
ACEI/ARB	45.0	43.0	48.5	44.8	44.9
Potassium-sparing diuretics†	2.1	1.4	2.0	0.97	1.7

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Table 1–Continued

	LAVI			
	Normal (≤34 mL/m²)	**Mild Enlargement (>34 to ≤41 mL/m²)**	**Moderate Enlargement (>41 to ≤48 mL/m²)**	**Severe Enlargement (>48 mL/m²)**
Participants, *n*	4356	400	107	54
Mean age (SD), *y*	75.1 (4.9)	76.2 (5.3)	77.1 (5.6)	77.9 (5.5)
Female sex, %	60.6	51.5	54.2	50.0
Black race, %	20.6	22.3	29.9	27.8
Medical conditions, %				
Heart failure	4.4	7.0	15.9	27.8
Hypertension	71.5	80.3	81.3	81.5
Myocardial infarction	7.1	12.3	9.4	24.1
Peripheral artery disease	8.5	10.3	15.9	18.5
Diabetes	29.8	31.0	31.8	22.2
Medications, %				
Aspirin	68.0	75.3	68.2	74.1
ACEI/ARB	44.0	53.3	51.4	46.3
Potassium-sparing diuretics†	1.7	0.75	1.9	3.7

ACEI = angiotensin-converting enzyme inhibitor; ARB = angiotensin-receptor blocker; LAVI = left atrial volume index.
* Categorical data are presented as percentages and continuous variables as means (SDs).
† Includes spironolactone and eplerenone.

Association of Left Atrial Echocardiographic Measures With Ischemic Stroke

Restricted cubic splines (Supplement Figure 3, available at Annals.org) showed that left atrial reservoir and conduit strain were linearly associated with ischemic stroke. By contrast, left atrial contractile strain and LAVI were not significantly associated with ischemic stroke.

The absolute difference in the cumulative incidence of ischemic stroke between the highest and lowest categories of the left atrial measures was largest for reservoir and conduit strain (Figure 2). Over 5 years, the cumulative incidences of ischemic stroke in the lowest quintiles of left atrial reservoir, conduit, and contractile strain were 2.99% (95% CI, 1.89% to 4.09%), 3.18% (CI, 2.14% to 4.22%), and 2.15% (CI, 1.09% to 3.21%), respectively. For the highest quintiles, these values were 1.42% (CI, 0.64% to 2.20%), 1.17 (CI, 0.44% to 1.90%), and 1.91% (CI, 0.91% to 2.91%), respectively. For severe left atrial enlargement, the cumulative incidence of ischemic stroke over 5 years was 1.99% (CI, 0.23% to 3.75%); for normal left atrial size, it was 1.71% (CI, 1.32% to 2.10%).

Ischemic Stroke Prediction

The ischemic stroke prediction model that consisted of left atrial reservoir strain and CHA_2DS_2-VASc variables–model 1A (reservoir strain)–had lower AIC than the benchmark model (model 0) and models that added left atrial conduit strain (model 1B), contractile strain (model 1C), or LAVI (model 1D) to model 0 (Table 2). Furthermore, addition of conduit strain (model 2), contractile strain (model 3), or LAVI (model 4) to model 1A (reservoir strain) resulted in higher AIC than that of model 1A (reservoir strain). Of the left atrial measures, only reservoir strain remained in the most parsimonious model generated using stepwise backward selection (Supplement Table 2, available at Annals.org).

Secondary and Sensitivity Analyses

In secondary and sensitivity analyses (Supplement Tables 3 to 9, available at Annals.org), the association of lower left atrial reservoir strain with ischemic stroke remained significant after adjustment for left atrial size, race–study center, medication use, incident AF or anticoagulation use, competing risk for death, and potential bias introduced by nonattendance at visit 5. Lower left atrial reservoir strain was associated with thrombotic stroke in stroke subtype analysis. This association was driven by an association with definite thrombotic nonlacunar stroke because none of the left atrial measures were associated with definite thrombotic lacunar stroke. Lower left atrial reservoir strain was also associated with noncarotid embolic stroke. Of note, LAVI was not associated with ischemic stroke in any of the sensitivity analyses after adjustment for left atrial reservoir strain and incident AF. Additional details are in the Supplement Results (available at Annals.org).

Risk Classification

The overall net improvement in risk classification by model 1A (reservoir strain) compared with model 0 was driven primarily by correct reclassification of participants who developed stroke to higher risk categories (Table 3). Of the 112 participants who developed stroke within 5 years, 13 (11.6% of those with stroke) were reclassified correctly to higher risk categories and 2 (1.8% of those with stroke) were reclassified incorrectly to lower risk categories. Of the 4805 participants who did not develop stroke at 5 years, 588 (12.2% of those without stroke) were reclassified correctly to lower risk categories and 612 (12.7% of those without stroke) were reclassified incorrectly to higher risk categories.

Decision Curve Analysis

Model 1A (reservoir strain) showed a greater net benefit than model 0, an intervention-for-all strategy,

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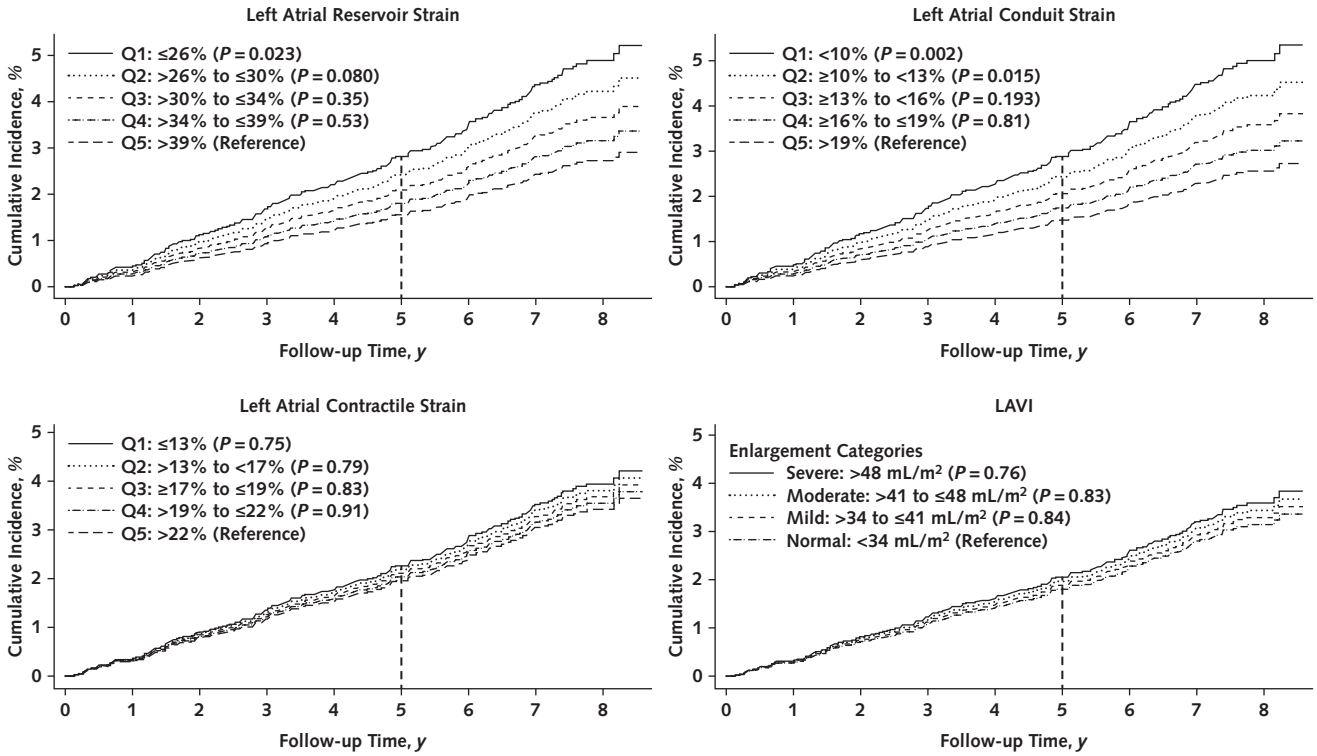
Figure 2. Cumulative incidence of stroke.



Cumulative incidence curves are adjusted for age, sex, race–study center, heart failure, hypertension, myocardial infarction, peripheral artery disease, and LAVI (left atrial reservoir strain for LAVI). Competing risk for death was accounted for. At 5 y (*vertical dashed line*), the cumulative incidence of stroke for each quintile (for left atrial strain) or category of enlargement (for LAVI) is as follows. Left atrial reservoir strain: Q1, 2.99% (95% CI, 1.89% to 4.09%); Q2, 2.57% (CI, 1.55% to 3.59%); Q3, 2.01% (CI, 1.07% to 2.95%); Q4, 1.79% (CI, 0.93% to 2.65%); and Q5, 1.42% (CI, 0.64% to 2.20%). Left atrial conduit strain: Q1, 3.18% (CI, 2.14% to 4.22%); Q2, 2.74% (CI, 1.70% to 3.78%); Q3, 1.99% (CI, 0.99% to 2.99%); Q4, 1.30% (CI, 0.56% to 2.04%); and Q5, 1.17% (CI, 0.44% to 1.90%). Left atrial contractile strain: Q1, 2.15% (CI, 1.09% to 3.21%); Q2, 2.11% (CI, 1.04% to 3.18%); Q3, 2.07% (CI, 1.01% to 3.13%); Q4, 1.99% (CI, 0.97% to 3.01%); and Q5, 1.91% (CI, 0.91% to 2.91%). LAVI: severe, 1.99% (CI, 0.23% to 3.75%); moderate, 1.91% (CI, 0.50% to 3.24%); mild, 1.79% (CI, 1.10% to 2.48%); and normal, 1.71% (CI, 1.32% to 2.10%). *P* values correspond to cumulative incidence at 5 y for each variable. LAVI = left atrial volume index; Q = quintile.

and an intervention-for-none strategy over absolute 5-year risk thresholds of 1% to 8% (**Figure 3**; **Supplement Table 10**, available at Annals.org). For example, at a 5-year risk threshold of 5% (that is, it is 19 times worse not to correctly predict stroke than to incorrectly predict it), the total true-positive classification minus the false-positive classification weighted by risk threshold (1/19) resulted in a predicted net benefit of 1.34 for every 1000 people to whom model 1A (reservoir strain) was applied and 0.36 for every 1000 people to whom model 0 was applied.

DISCUSSION

In a prospective, community-based cohort of Black and White older adults without prior AF or stroke in the United States, left atrial mechanical dysfunction, detected by analysis of left atrial strain, was associated with ischemic stroke independently of left atrial size and CHA_2DS_2-VASc variables. The addition of left atrial reservoir strain to the CHA_2DS_2-VASc variables produced the best-quality prediction model with the lowest prediction error, improved risk classification, and yielded a greater predicted net benefit, as shown by decision curve analysis.

Fibrotic myocardial remodeling is the mechanism underlying the structural and functional changes that characterize atrial myopathy (2). It can occur in response to stressors, including pressure and volume overload, ischemia, inflammation, oxidative stress, and aging (5). Atrial mechanical dysfunction can lead to blood stasis, which is a key mechanism linking atrial myopathy to cardiac embolism (40). Lower left atrial strain has been associated with a higher degree of left atrial fibrosis (41), as well as a lower velocity of left atrial appendage emptying and a higher prevalence of left atrial appendage thrombi (42). In the present study, lower left atrial strain was significantly associated with nonlacunar and noncarotid embolic stroke, but not with lacunar stroke. Of the thrombotic stroke subtypes, nonlacunar strokes are more likely than smaller deep-brain lacunar infarcts—which are postulated to be secondary to intrinsic cerebral arteriolar abnormalities—to be related to thromboembolism (43). In aggregate, the results of this study support the hypothesis that atrial myopathy, characterized by left atrial mechanical dysfunction, is

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Table 2. Association of Left Atrial Function and Size With Ischemic Stroke and Evaluation of Model Prediction Error*

Variable	Model 1	Model 2	Model 3	Model 4
HR (95% CI) for left atrial measures				
Reservoir strain	A) 1.27 (1.14–1.41)	1.23 (1.08–1.40)	1.32 (1.14–1.54)	1.25 (1.12–1.40)
Conduit strain	B) 1.34 (1.13–1.58)	1.09 (0.88–1.35)	–	–
Contractile strain	C) 1.23 (1.05–1.43)	–	0.92 (0.74–1.14)	–
Volume index	D) 1.09 (1.02–1.16)	–	–	1.03 (0.96–1.11)
Model AIC	†	2876.19	2876.24	2876.24

AIC = Akaike information criterion; HR = hazard ratio.
* Cox proportional hazards models. Left atrial measures were modeled per 5-percentage point decrement (strain) and per 5-mL/m^2 increment (volume index). Model 0 (benchmark): age, sex, heart failure, hypertension, diabetes, peripheral artery disease, and myocardial infarction (AIC = 2892.85). Model 1: model 0 + one left atrial measure. Model 2: model 0 + left atrial reservoir strain and left atrial conduit strain. Model 3: model 0 + left atrial reservoir strain and left atrial contractile strain. Model 4: model 0 + left atrial reservoir strain and left atrial volume index.
† AICs for each version of model 1 are 2874.75, 2883.39, 2887.81, and 2890.43 for models 1A (reservoir strain), 1B (conduit strain), 1C (contractile strain), and 1D (volume index), respectively.

intrinsically prothrombotic, resulting in higher risk for cardiac embolism and ischemic stroke.

Although increased left atrial size has traditionally been considered to be a marker of atrial remodeling, it was not independently associated with ischemic stroke and did not improve stroke prediction in the present study. Similarly, recent analyses from MESA (Multi-Ethnic Study of Atherosclerosis) (44) and CHS (Cardiovascular Health Study) (45) did not find a significant relationship between left atrial size and stroke risk. By contrast, the association of lower echocardiographic left atrial strain with higher risk for ischemic or embolic stroke has been previously reported in cross-sectional (12), case–control (13), and retrospective cohort (14, 15) studies of patients with stroke or AF. Critical knowledge gaps remain that prevent translation of these data to clinical practice. First, there are limited analyses from prospective studies enrolling low-risk (without prevalent AF or stroke) participants from the general population with rigorous adjudication of ischemic stroke. Second, whether analysis of left atrial strain can improve prediction of ischemic stroke in people without prior AF and stroke is unclear.

The present study specifically addresses the foregoing gaps and advances the field by showing that, of the left atrial measures studied, reservoir strain was the most important marker for ischemic stroke prediction in people without prior AF or stroke. The improvement in stroke prediction models attributed to left atrial reservoir strain was reflected in improved risk classification, and clinical utility was shown in predicted net benefit, which weighs the potential harms and benefits of using the model to aid clinical decisions. Of note, 2-dimensional echocardiography was used to evaluate left atrial function and size. Compared with other cardiac imaging techniques, such as cardiac magnetic resonance imaging, echocardiography is a cost-effective, ubiquitous tool, making it a practical choice for widespread clinical use.

There are some limitations to consider. First, in 2019, ARIC staff could not contact 19% of participants. However, ascertainment of outcome events in ARIC is based not only on information obtained from contact with participants or proxies but also on systematic data abstraction from regional hospitalization records in each of the communities. Therefore, outcome events might have been missed only for participants with whom ARIC had lost

Table 3. Improvement in Classification of Observed Ischemic Stroke Events After 5 Years by Addition of Left Atrial Reservoir Strain to CHA_2DS_2-VASc Variables

Model 0*	Model 1 (Reservoir Strain)†			
	<2.5%	2.5%–5%	>5%	Total
Participants with ischemic stroke				
<2.5%	90	8‡	0‡	98
2.5%–5%	2§	7	5‡	14
>5%	0§	0§	0	0
Total	92	15	5	112
Participants without ischemic stroke				
<2.5%	2536	482§	8§	3026
2.5%–5%	547‡	1021	122§	1690
>5%	1‡	40‡	48	89
Total	3084	1543	178	4805

* Model 0: age, sex, hypertension, myocardial infarction, peripheral artery disease, heart failure, and diabetes.
† Model 1 (reservoir strain): model 0 + left atrial reservoir strain. Net reclassification improvement: 0.10 (95% CI, 0.03–0.17). Integrated discrimination improvement: 0.61 (95% CI, 0.50–0.72).
‡ Favorable reclassification of observed events.
§ Unfavorable reclassification of observed events.

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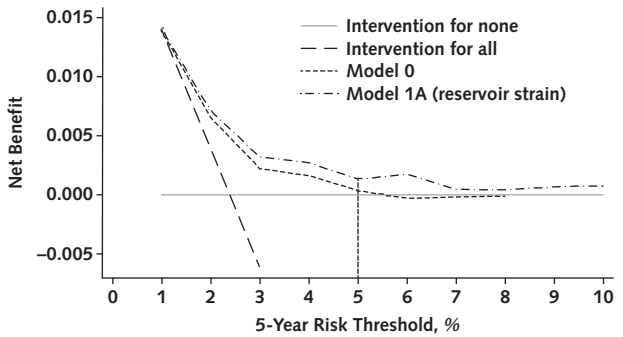
Figure 3. Decision curves of ischemic stroke prediction models.



Model 0 contains age, sex, heart failure, diabetes, hypertension, myocardial infarction, and peripheral artery disease. Model 1A (reservoir strain) is model 0 + left atrial reservoir strain. At an absolute 5-y risk threshold of 5% (*vertical dashed line*), the predicted net benefit (true positive – weighted false positive) was 1.34 per 1000 people with model 1 (reservoir strain) and 0.36 per 1000 people with model 0.

contact and who were hospitalized outside the region in which they were enrolled. Second, whether the results of the present study are generalizable to younger people is unclear. However, because atrial myopathy and ischemic stroke are more prevalent in older adults, the findings of this study have significant public health relevance. Third, because AF was identified primarily from hospital discharge records and death certificates, subclinical AF and AF managed exclusively in the outpatient setting were not accounted for. However, the incidence of AF in the ARIC study is consistent with that in other population-based studies, and using hospital discharge records for the purposes of AF detection has been previously validated (30, 46–48). Fourth, although the use of 2-week continuous ambulatory cardiac monitoring during visit 6 would help identify subclinical AF, only 2616 participants had ambulatory cardiac monitoring. Finally, although statistical adjustment was made for potential confounders, imperfectly measured and unmeasured variables were not accounted for—as is the case in all observational studies.

Of note, stroke can be the first clinical manifestation of atrial myopathy before development or detection of AF (49). The findings of the present study provide the premise for future clinical trials designed to evaluate primary prevention of stroke in older adults with atrial myopathy but without AF or prior stroke. To this end, external validation with analyses evaluating prediction model discrimination and calibration along with formal harm–benefit analyses for potential interventions (such as anticoagulation) will be needed. In addition, because patients with AF also have varying degrees of atrial myopathy, future research is warranted to quantify the extent to which the risk for ischemic stroke in patients with AF is attributed to the dysrhythmia of AF per se versus the underlying atrial myopathy.

In conclusion, in people without AF or prior stroke, lower left atrial reservoir strain is independently associated with and can aid in prediction of ischemic stroke. External validation and formal decision analyses are needed before these data can aid in the design of clinical trials for stroke prevention.

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Reproducible Research Statement: *Study protocol*: Study manuals for the ARIC study are publicly available on the ARIC study website (https://sites.cscc.unc.edu/aric/cohort-manuals). *Statistical code*: Available in the **Supplement** (available at Annals.org). *Data set:* Data can be requested by contacting the ARIC Coordinating Center. Details on data availability are in the **Supplement.**

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